The Hong Kong University of Science and Technology
IEDA 1250, Summer 2026
Mid-term Exam Solution

Student ID ________________ Name: ________________________________

Please read the following instructions carefully.

- You have **90 minutes** to complete this exam. This question booklet contains 6 questions (including a bonus question) for the total of 120 points. Check to see if any pages are missing.
- Read the instructions for individual questions carefully before answering the questions.

Question	Points	Score
1	10	
2	20	
3	20	
4	20	
5	20	
6	30	
Total	120	

1. **(10 points)** Solve the following linear program (LP) graphically. Clearly identify the feasible region, the optimal solution, and the optimal objective value.

$$\begin{aligned} \max_{x_1,x_2} \quad & x_1 + \frac{1}{2}x_2 \\ \text{s.t.} \quad & x_1 + 2x_2 \le 6, \\ & x_1 + 2x_2 \ge 0, \\ & 2x_1 - x_2 \le 4, \\ & 2x_1 - x_2 \ge -2. \end{aligned}$$

Solution: The feasible region is a diamond with vertices

$$\left(-\frac{4}{5}, \frac{2}{5}\right), \quad \left(\frac{2}{5}, \frac{14}{5}\right), \quad \left(\frac{14}{5}, \frac{8}{5}\right), \quad \left(\frac{8}{5}, -\frac{4}{5}\right).$$

Now evaluate the objective function

$$z = x_1 + \frac{1}{2}x_2$$

at the four vertices:

Vertex	$z = x_1 + \frac{1}{2}x_2$
$\left(-\frac{4}{5}, \frac{2}{5}\right)$	$-\frac{3}{5}$
$\left(\frac{8}{5}, -\frac{4}{5}\right)$	$\frac{7}{5}$
$\left(\frac{2}{5}, \frac{14}{5}\right)$	$\frac{9}{5}$
$\left(\frac{14}{5}, \frac{8}{5}\right)$	$\frac{18}{5}$

Therefore, the optimal solution is

$$\boxed{x_1^* = \frac{14}{5}, \qquad x_2^* = \frac{8}{5}}$$

and the optimal objective value is

$$\boxed{z^* = \frac{18}{5}.}$$

2. **(20 points)** Consider the following primal LP:

$$\begin{aligned} \max \quad & 2x_1 + 3x_2 + x_3 + x_4 + x_5 \\ \text{s.t.} \quad & 2x_1 + x_2 + x_3 + x_4 \le 3, \\ & x_1 - x_2 + 2x_3 + x_5 = 7, \\ & x_1 + 2x_2 - x_3 + 3x_4 + x_5 \le 6, \\ & x_2 + x_3 - x_4 + 2x_5 = 7, \\ & x_1 \ge 0, \quad x_2 \ge 0, \end{aligned}$$

where x_3, x_4, x_5 are unrestricted in sign. Suppose that y_1, y_2, y_3, and y_4 are the dual variables corresponding to the four constraints of the primal LP, respectively, and that the optimal dual solution is

$$(y_1^*, y_2^*, y_3^*, y_4^*) = (2, -1, 0, 1).$$

Use the complementary slackness conditions to find an optimal primal solution.

Solution: Given the primal LP and the optimal dual solution

$$(y_1^*, y_2^*, y_3^*, y_4^*) = (2, -1, 0, 1),$$

we use complementary slackness to find an optimal primal solution.

The dual LP is

$$\begin{aligned}
\min \quad & 3y_1 + 7y_2 + 6y_3 + 7y_4 \\
\text{s.t.} \quad & 2y_1 + y_2 + y_3 \geq 2, \\
& y_1 - y_2 + 2y_3 + y_4 \geq 3, \\
& y_1 + 2y_2 - y_3 + y_4 = 1, \\
& y_1 + 3y_3 - y_4 = 1, \\
& y_2 + y_3 + 2y_4 = 1, \\
& y_1 \geq 0, \quad y_3 \geq 0, \quad y_2, y_4 \text{ free.}
\end{aligned}$$

Step 1: Apply complementary slackness.

For primal constraints:

$$y_1^* \left(3 - (2x_1 + x_2 + x_3 + x_4)\right) = 0.$$

Since $y_1^* = 2 \neq 0$, the first primal constraint must be tight:

$$2x_1 + x_2 + x_3 + x_4 = 3.$$

Also,

$$y_3^* \left(6 - (x_1 + 2x_2 - x_3 + 3x_4 + x_5)\right) = 0.$$

Since $y_3^* = 0$, this gives no additional equality condition.

For dual constraints corresponding to x_1 and x_2:

$$x_1 \left(2y_1^* + y_2^* + y_3^* - 2\right) = 0.$$

Substituting the dual solution gives

$$2(2) + (-1) + 0 - 2 = 1 \neq 0.$$

Therefore,

$$x_1 = 0.$$

Similarly,

$$x_2 \left(y_1^* - y_2^* + 2y_3^* + y_4^* - 3\right) = 0.$$

Substituting the dual solution gives

$$2 - (-1) + 2(0) + 1 - 3 = 1 \neq 0.$$

Therefore,

$$x_2 = 0.$$

Step 2: Substitute $x_1 = 0$ and $x_2 = 0$ into the primal constraints.

The tight first primal constraint gives

$$x_3 + x_4 = 3.$$

The second primal constraint gives

$$2x_3 + x_5 = 7.$$

The fourth primal constraint gives

$$x_3 - x_4 + 2x_5 = 7.$$

Therefore, we solve

$$\begin{cases} x_3 + x_4 = 3, \\ 2x_3 + x_5 = 7, \\ x_3 - x_4 + 2x_5 = 7. \end{cases}$$

Solving this system, we obtain

$$x_3 = 2, \qquad x_4 = 1, \qquad x_5 = 3.$$

Hence, an optimal primal solution is

$$\boxed{x_1^* = 0, \quad x_2^* = 0, \quad x_3^* = 2, \quad x_4^* = 1, \quad x_5^* = 3.}$$

The optimal primal objective value is

$$2x_1^* + 3x_2^* + x_3^* + x_4^* + x_5^* = 0 + 0 + 2 + 1 + 3 = 6.$$

Thus,

$$\boxed{z^* = 6.}$$

3. **(20 points)** A relief agency needs to ship medicine boxes from two warehouses to three clinics. Warehouse 1 has 20 boxes available, and Warehouse 2 has 30 boxes available. Clinic 1 needs 10 boxes, Clinic 2 needs 25 boxes, and Clinic 3 needs 15 boxes. Since total supply equals total demand, all boxes must be shipped. The shipping cost per box is shown below.

	Clinic 1	Clinic 2	Clinic 3
Warehouse 1	4	6	8
Warehouse 2	5	4	3

Let x_{ij} denote the number of boxes shipped from Warehouse i to Clinic j.

(a) Formulate a linear program that minimizes the total shipping cost while satisfying all clinic demands.

(b) Write down the dual linear program. Let u_1, u_2 be the dual variables associated with the two warehouse constraints, and let v_1, v_2, v_3 be the dual variables associated with the three clinic constraints.

(c) Suppose the optimal shipment plan is

$$x_{11}^* = 10, \quad x_{12}^* = 10, \quad x_{13}^* = 0, \quad x_{21}^* = 0, \quad x_{22}^* = 15, \quad x_{23}^* = 15.$$

Also suppose an optimal dual solution is

$$u_1^* = 2, \quad u_2^* = 0, \quad v_1^* = 2, \quad v_2^* = 4, \quad v_3^* = 3.$$

Explain the meaning of the dual variables intuitively. What is the economic interpretation of the dual LP and its optimal solution?

Solution: Let x_{ij} denote the number of boxes shipped from Warehouse i to Clinic j.

(a) Primal LP formulation.

The objective is to minimize the total shipping cost:

$$\min \quad 4x_{11} + 6x_{12} + 8x_{13} + 5x_{21} + 4x_{22} + 3x_{23}.$$

Subject to the warehouse supply constraints:

$$x_{11} + x_{12} + x_{13} = 20,$$

$$x_{21} + x_{22} + x_{23} = 30,$$

and the clinic demand constraints:

$$x_{11} + x_{21} = 10,$$

$$x_{12} + x_{22} = 25,$$

$$x_{13} + x_{23} = 15.$$

Also,

$$x_{ij} \geq 0, \qquad i = 1, 2, \quad j = 1, 2, 3.$$

(b) Dual LP formulation.

Let u_1, u_2 be the dual variables associated with the two warehouse constraints, and let v_1, v_2, v_3 be the dual variables associated with the three clinic constraints. Since all primal constraints are equality constraints, all dual variables are unrestricted in sign.

The dual LP is

$$\max \quad 20u_1 + 30u_2 + 10v_1 + 25v_2 + 15v_3$$

subject to

$$u_1 + v_1 \leq 4,$$

$$u_1 + v_2 \leq 6,$$

$$u_1 + v_3 \leq 8,$$

$$u_2 + v_1 \leq 5,$$

$$u_2 + v_2 \leq 4,$$

$$u_2 + v_3 \leq 3,$$

where

$$u_1, u_2, v_1, v_2, v_3 \text{ are free.}$$

(c) Economic interpretation of the dual variables.

The dual variables can be interpreted as shadow prices. The variables u_1 and u_2 represent the marginal values associated with the supplies available at Warehouses 1 and 2, respectively. The variables v_1, v_2, v_3 represent the marginal values associated with satisfying demands at Clinics 1, 2, and 3, respectively.

The dual constraint

$$u_i + v_j \leq c_{ij}$$

means that the combined value of one box supplied from Warehouse i and delivered to Clinic j cannot exceed the actual shipping cost c_{ij}. Therefore, the dual LP tries to assign values to warehouse supplies and clinic demands so that these values are consistent with all shipping costs.

Using the given optimal dual solution

$$u_1^* = 2, \qquad u_2^* = 0, \qquad v_1^* = 2, \qquad v_2^* = 4, \qquad v_3^* = 3,$$

the dual objective value is

$$20(2) + 30(0) + 10(2) + 25(4) + 15(3) = 40 + 0 + 20 + 100 + 45 = 205.$$

The cost of the given primal shipment plan is

$$4(10) + 6(10) + 8(0) + 5(0) + 4(15) + 3(15) = 40 + 60 + 0 + 0 + 60 + 45 = 205.$$

Since the primal and dual objective values are equal, the given shipment plan and the given dual solution are both optimal.

Economically, the optimal dual solution provides a system of prices that supports the optimal transportation plan. Routes with positive shipments satisfy

$$u_i^* + v_j^* = c_{ij},$$

while routes with zero shipments satisfy

$$u_i^* + v_j^* \leq c_{ij}.$$

Thus, only routes whose shipping cost is justified by the corresponding warehouse and clinic shadow prices are used in the optimal plan.

4. **(20 points)** A manufacturer produces a single product over the next 6 months. The demand in each month is known and must be satisfied on time. Shortages are not allowed. The manufacturer may produce in a month only if it sets up the production line in that month. If the production line is set up in month t, where $t \in \{1, 2, \ldots, 6\}$, the firm pays a fixed setup cost. In addition, it pays a variable production cost for each unit produced and an inventory holding cost for each unit carried from one month to the next.

The monthly data are shown below.

Month	1	2	3	4	5	6
Demand	40	60	35	70	55	45
Production capacity	80	80	80	80	80	80
Variable production cost	12	11	13	10	12	14
Setup cost	300	300	300	300	300	300
Holding cost	2	2	2	2	2	–

Assume that the initial inventory is zero and that the company wants no inventory left over at the end of Month 6. Formulate this problem as a mixed-integer linear program that minimizes the total production, setup, and inventory holding cost. Clearly define all decision variables.

Solution: Let x_t be the number of units produced in month t, I_t be the inventory at the end of month t, and

$$y_t = \begin{cases} 1, & \text{if the production line is set up in month } t, \\ 0, & \text{otherwise,} \end{cases} \qquad t = 1, \ldots, 6.$$

The mixed-integer linear program is

$$\begin{aligned}\min \quad & 12x_1 + 11x_2 + 13x_3 + 10x_4 + 12x_5 + 14x_6 \\ & + 300(y_1 + y_2 + y_3 + y_4 + y_5 + y_6) \\ & + 2(I_1 + I_2 + I_3 + I_4 + I_5)\end{aligned}$$

subject to

$$x_1 = 40 + I_1,$$

$$I_1 + x_2 = 60 + I_2, \qquad I_2 + x_3 = 35 + I_3,$$

$$I_3 + x_4 = 70 + I_4, \qquad I_4 + x_5 = 55 + I_5,$$

$$I_5 + x_6 = 45 + I_6,$$

$$I_6 = 0,$$

$$0 \le x_t \le 80y_t, \qquad t = 1, \dots, 6,$$

$$I_t \ge 0, \qquad t = 1, \dots, 6,$$

and

$$y_t \in \{0, 1\}, \qquad t = 1, \dots, 6.$$

The constraints $x_t \le 80y_t$ ensure that production in month t is possible only if the setup decision $y_t = 1$.

5. **(20 points)** Consider a two-player zero-sum game where Player A's payoffs are shown below. Player A chooses rows ($A1$ to $A5$), and Player B chooses columns ($B1$ to $B5$).

Player A\Player B	$B1$	$B2$	$B3$	$B4$	$B5$
$A1$	4	1	5	2	6
$A2$	2	3	4	4	5
$A3$	1	0	2	1	3
$A4$	1	2	3	3	4
$A5$	0	0	1	1	2

Answer the following questions:

(a) Eliminate all dominated strategies for Player A and Player B.

(b) Does the reduced game have a pure strategy Nash equilibrium? If yes, find the equilibrium; if no, explain why every payoff in the reduced game is unstable.

Solution: Since this is a zero-sum game, Player A maximizes the payoff and Player B minimizes Player A's payoff.

(a) Elimination of dominated strategies. For Player A, Row $A3$ is dominated by Row $A1$, Row $A4$ is dominated by Row $A2$, and Row $A5$ is dominated by Row $A1$. Therefore, Rows $A3, A4, A5$ can be eliminated.

For Player B, Column $B5$ is dominated by Column $B3$, Column $B3$ is dominated by Column $B2$, and Column $B4$ is dominated by Column $B2$. Therefore, Columns $B3, B4, B5$ can be eliminated.

The reduced game is

	$B1$	$B2$
$A1$	4	1
$A2$	2	3

(b) Pure strategy Nash equilibrium. The row minima are

$$\min(A1) = 1, \qquad \min(A2) = 2,$$

so $\max\min = 2$. The column maxima are

$$\max(B1) = 4, \qquad \max(B2) = 3,$$

so $\min\max = 3$. Since

$$\max\min = 2 \neq 3 = \min\max,$$

there is no pure strategy Nash equilibrium.

Equivalently, every payoff in the reduced game is unstable:

- At $(A1, B1)$, Player B can switch to $B2$ and reduce A's payoff from 4 to 1.
- At $(A1, B2)$, Player A can switch to $A2$ and increase the payoff from 1 to 3.
- At $(A2, B1)$, Player A can switch to $A1$ and increase the payoff from 2 to 4.
- At $(A2, B2)$, Player B can switch to $B1$ and reduce A's payoff from 3 to 2.

Therefore, the reduced game has no pure strategy Nash equilibrium.

6. **(30 points)** A firm has options to invest in 10 different projects. The total investment cannot exceed the available budget q. The profits and costs of these projects are given by $(p_1, p_2, \ldots, p_{10})$ and $(c_1, c_2, \ldots, c_{10})$, respectively. The firm wants to maximize its total profit. In addition to the budget constraint, the managers need to consider the following requirements:

   (a) Projects 2 and 5 cannot be chosen together.
   (b) Exactly 4 projects are to be selected.
   (c) At least one project must be selected from the project set $\{1, 3, 4\}$.
   (d) If Project 6 is selected, then Project 7 must also be selected.
   (e) If Project 8 is selected, then at least one of Projects 3 and 4 must also be selected.
   (f) Either both Projects 1 and 10 are selected, or neither of them is selected.
   (g) If both Projects 3 and 4 are selected, then Project 9 must also be selected.
   (h) Project 5 cannot be selected unless at least two projects from the set $\{6, 7, 8\}$ are selected.
   (i) At least one of the following two conditions must hold: at least two projects from the set $\{1, 2, 3, 4\}$ are selected, or at most one project from the set $\{5, 6, 7, 8\}$ is selected.
   (j) If Project 10 is selected, then at least one of the following two conditions must hold:
      - Both Projects 2 and 9 are selected.
      - Neither Project 3 nor Project 4 is selected.

   Define integer decision variables and formulate logical constraints for (a)–(j).

   Solution: Let

$$x_i = \begin{cases} 1, & \text{if Project } i \text{ is selected,} \\ 0, & \text{otherwise,} \end{cases} \qquad i = 1, \ldots, 10.$$

   The objective is

$$\max \quad \sum_{i=1}^{10} p_i x_i.$$

The budget constraint is

$$\sum_{i=1}^{10} c_i x_i \le q,$$

with

$$x_i \in \{0,1\}, \qquad i = 1, \dots, 10.$$

The logical constraints are

$$x_2 + x_5 \le 1,$$

$$\sum_{i=1}^{10} x_i = 4,$$

$$x_1 + x_3 + x_4 \ge 1,$$

$$x_6 \le x_7,$$

$$x_8 \le x_3 + x_4,$$

$$x_1 = x_{10},$$

$$x_9 \ge x_3 + x_4 - 1,$$

$$x_6 + x_7 + x_8 \ge 2x_5.$$

For part (i), introduce a binary variable $z_i \in \{0,1\}$ and impose

$$x_1 + x_2 + x_3 + x_4 \ge 2z_i,$$

$$x_5 + x_6 + x_7 + x_8 \le 1 + 3z_i.$$

If $z_i = 1$, the first condition is enforced. If $z_i = 0$, the second condition is enforced.

For part (j), introduce a binary variable $z_j \in \{0,1\}$ and impose

$$x_2 \ge x_{10} + z_j - 1,$$

$$x_9 \ge x_{10} + z_j - 1,$$

$$x_3 \le 1 - x_{10} + z_j,$$

$$x_4 \le 1 - x_{10} + z_j.$$

If $x_{10} = 1$ and $z_j = 1$, then $x_2 = x_9 = 1$. If $x_{10} = 1$ and $z_j = 0$, then $x_3 = x_4 = 0$. If $x_{10} = 0$, these constraints impose no additional restriction.